This study evaluates the impact of a mental health and academic achievement intervention across several schools. Using a multi-method analytical approach—including **Wilcoxon Signed-Rank tests**, **Chi-Square independence tests**, and **Spearman correlations**—we analyzed student response trends from PRE to POST phases. Results show a consistent, statistically significant improvement in competencies, particularly in **Connection** and **Procrastination**.

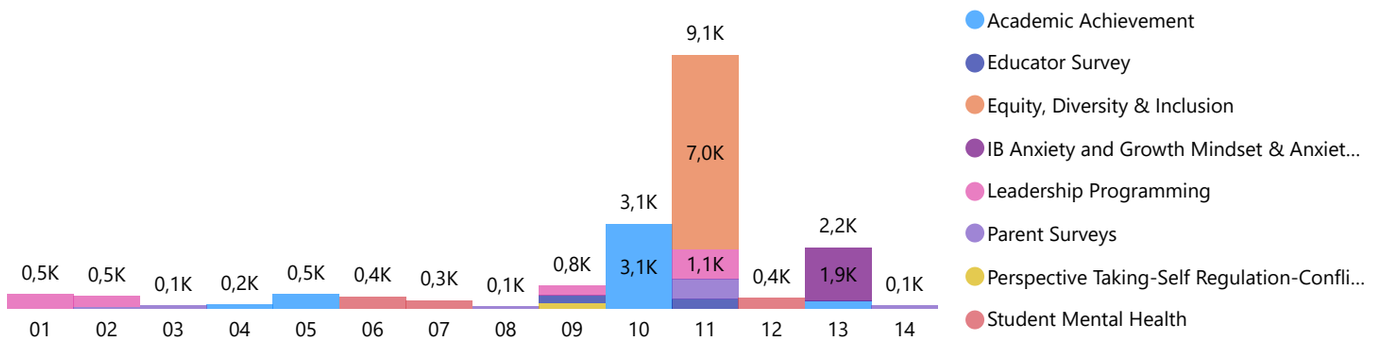
A **Random Forest Regressor** identified curriculum content and intervention phase as the primary drivers of student well-being (MAE=0.78). These findings validate the program's effectiveness, proving that structured mental health support builds measurable, scalable resilience in academic environments.

Team EMPI

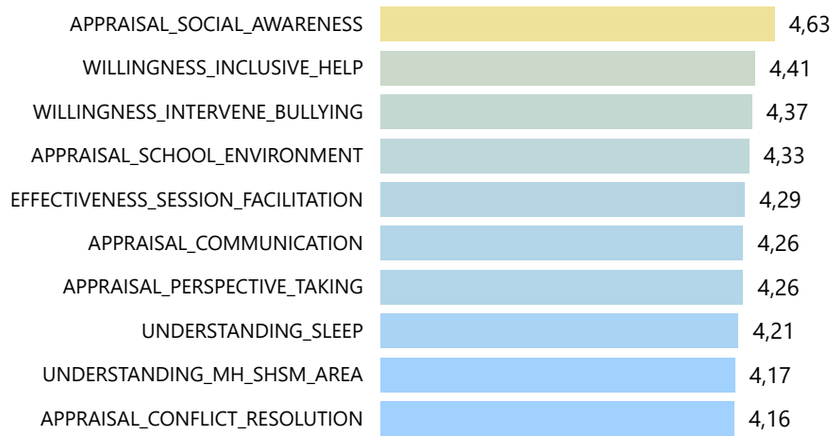
Gabriel Mantilla
Nathaniel Badinas
Lorena Lopez
Anjana Rathinam

Year: All
Month: All
Phase: All
Concept: All
Survey: All
School: All
Questions: All

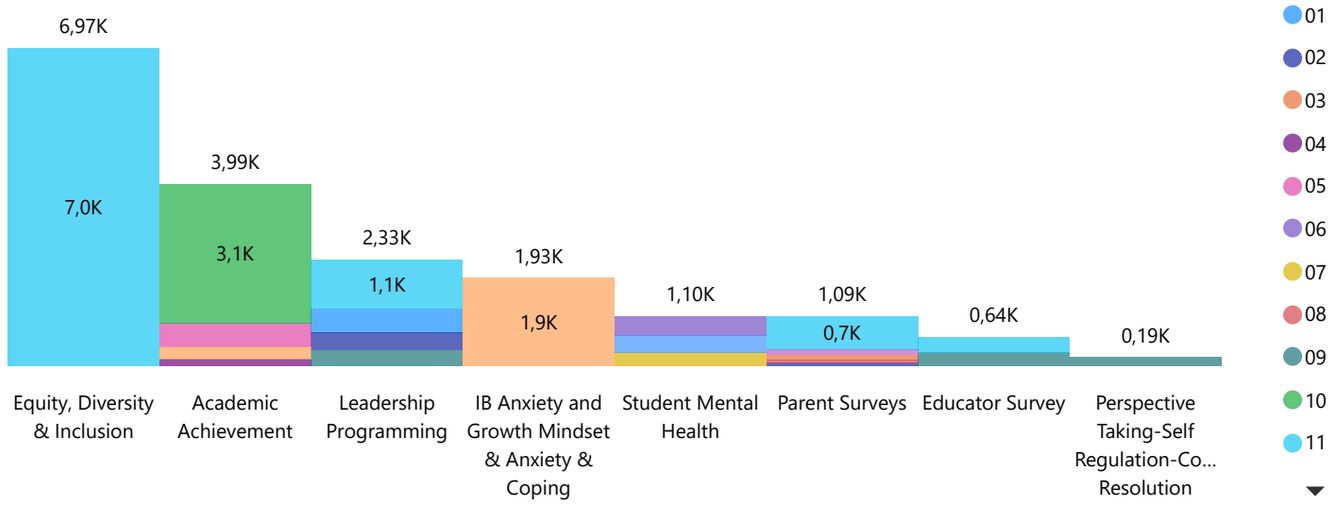
Frequency of School by Survey



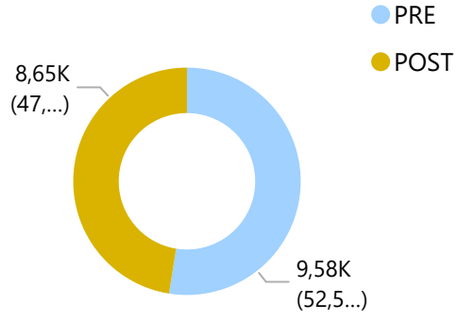
Top 10 of Concepts By Score



Frequency of Survey by Schools



Frequency of Phase

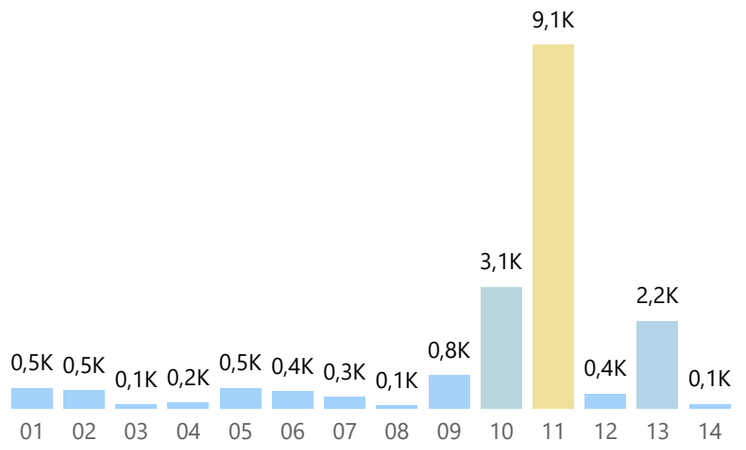


1909
Surveyed

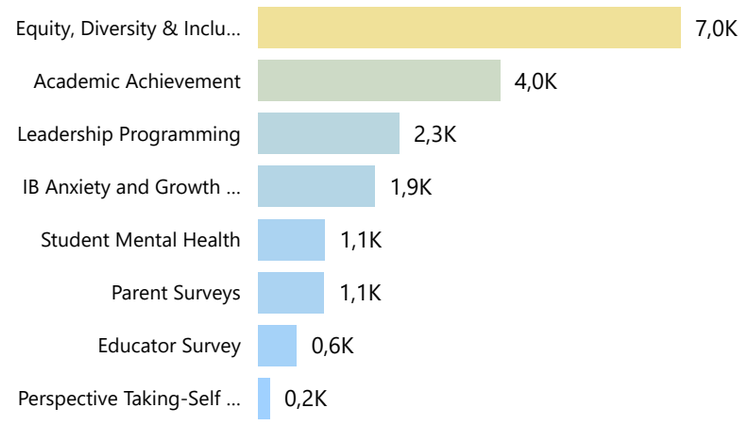
18,23K
Responses

Year: All
Month: All
Phase: All
Concept: All
Survey: All
School: All
Questions: All

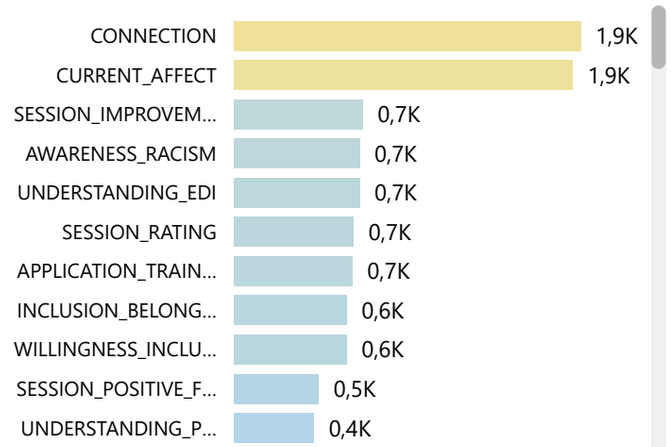
Frequency of Schools



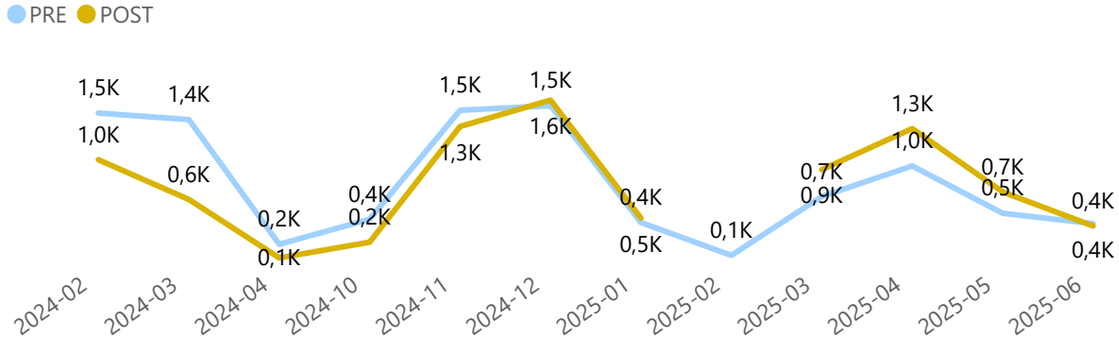
Frequency of Surveys



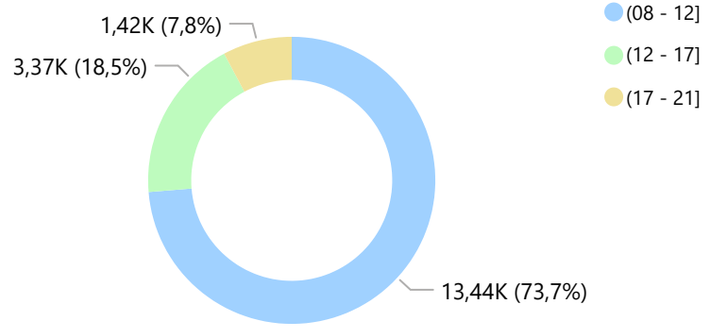
Frequency of Concepts



Survey Phase Trends



Hours Ranges

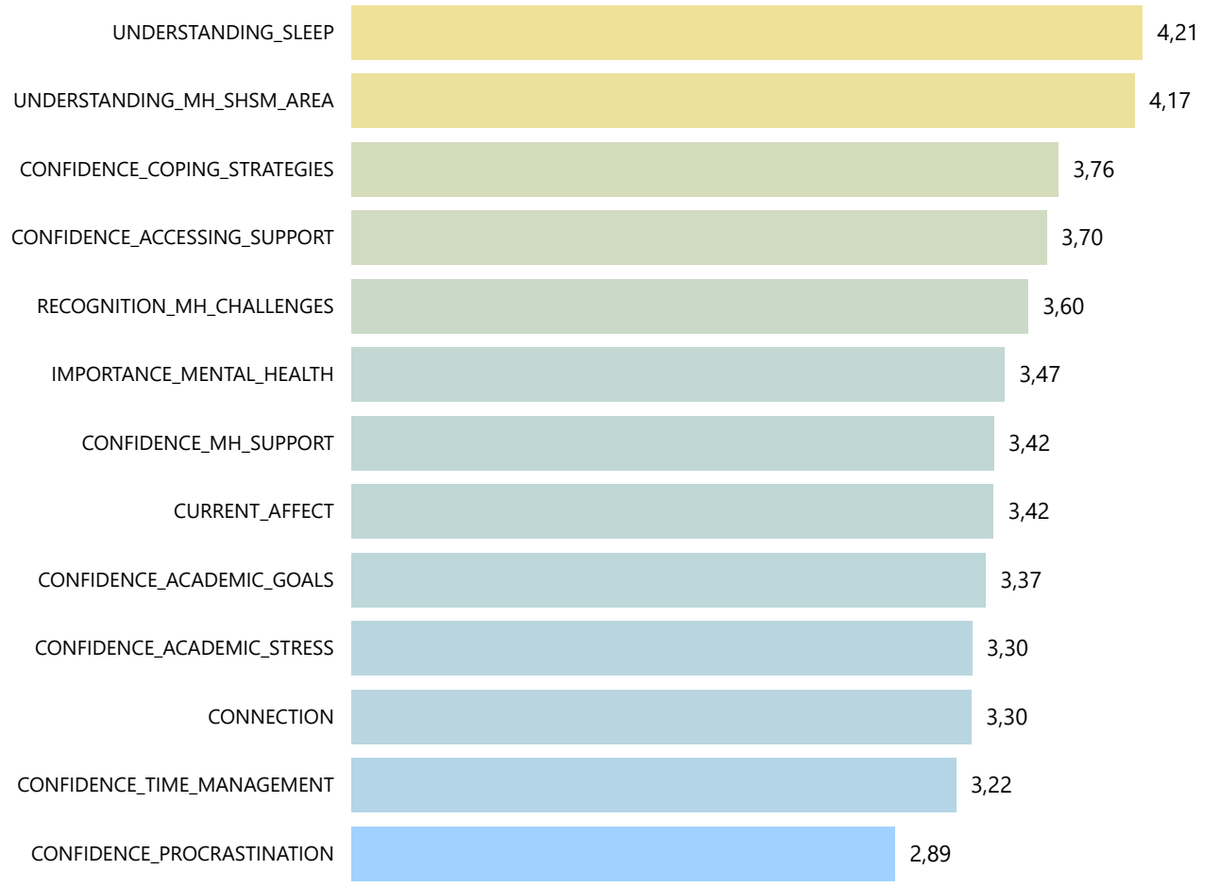


1909
Surveyed

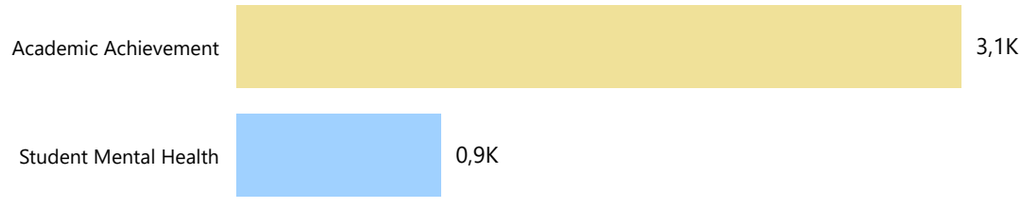
18,23K
Responses

Year	Month	Hour	Phase	Concept	Survey	School	Questions
All	All	All	All	All	All	All	All

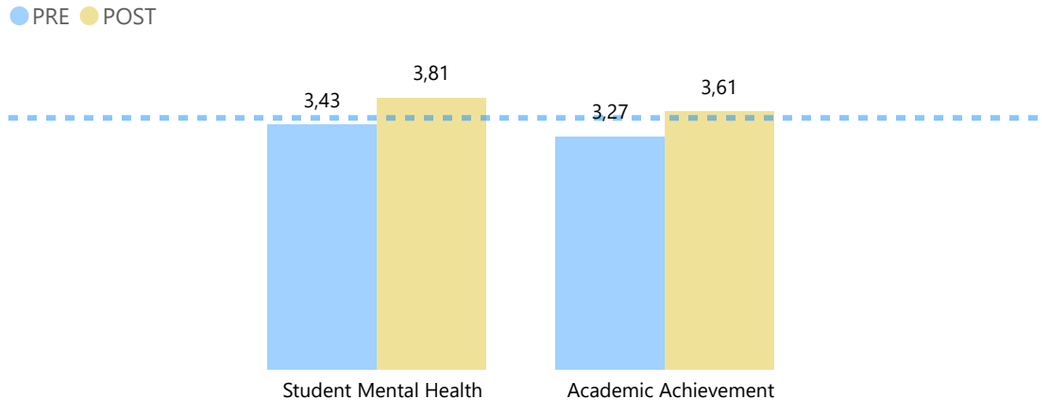
Comparision Concept by Phase



Frequency of Surveys

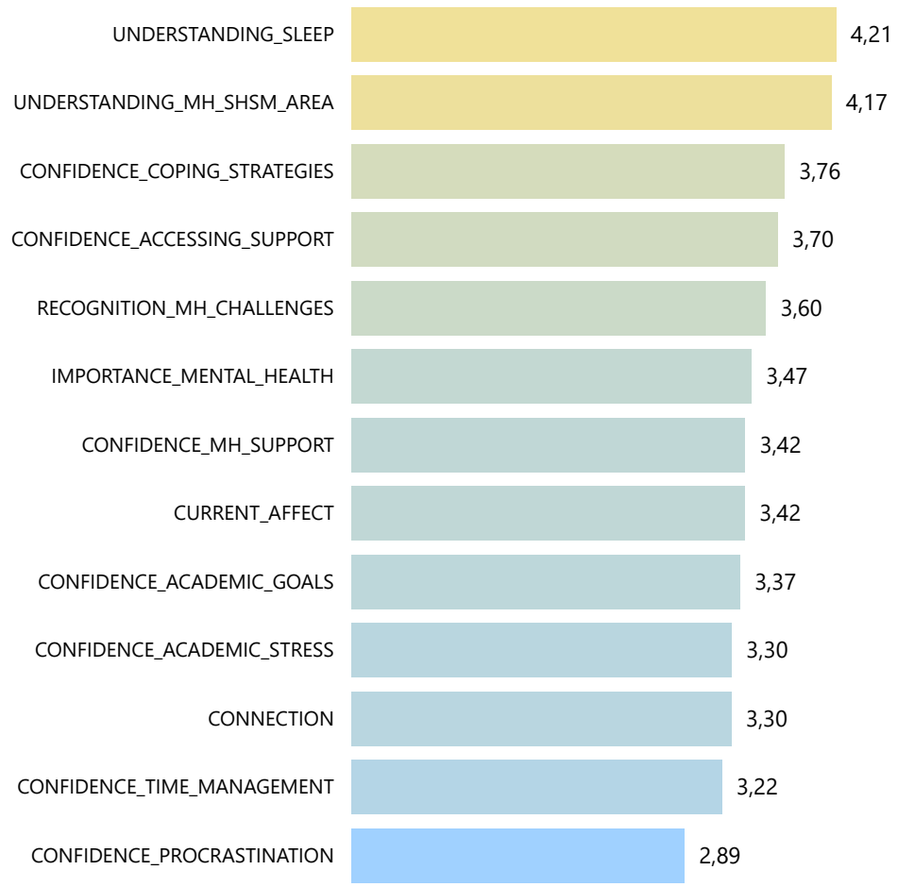


Comparision Response by Phase

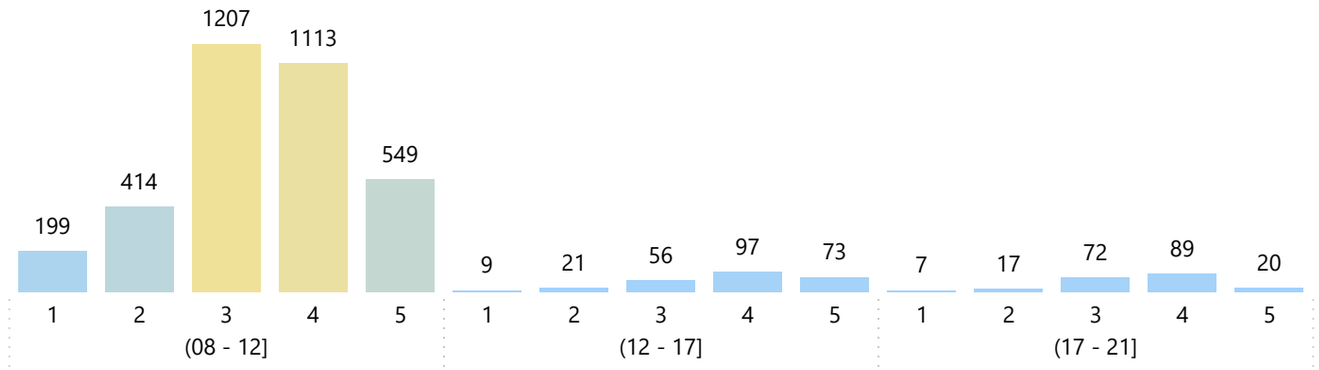


Year	Month	Hour	Phase	Concept	Survey	School	Questions
All	All	All	All	All	All	All	All

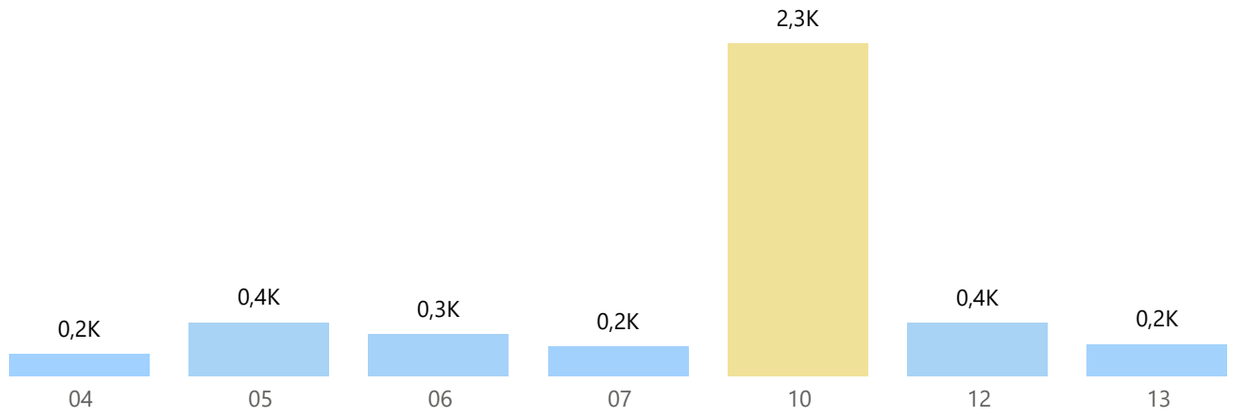
Comparision Concept by Phase



Hours Range ~ Responses

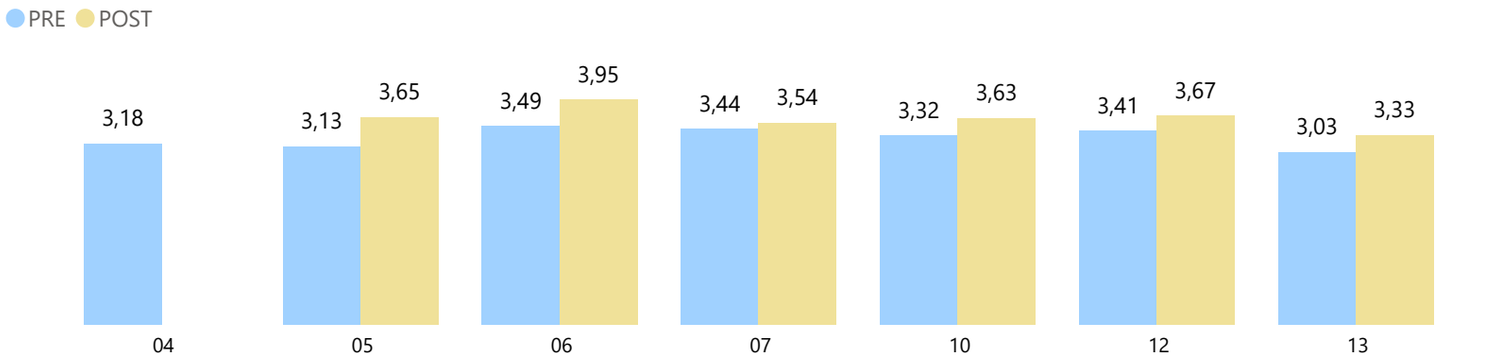


Frequency of Schools

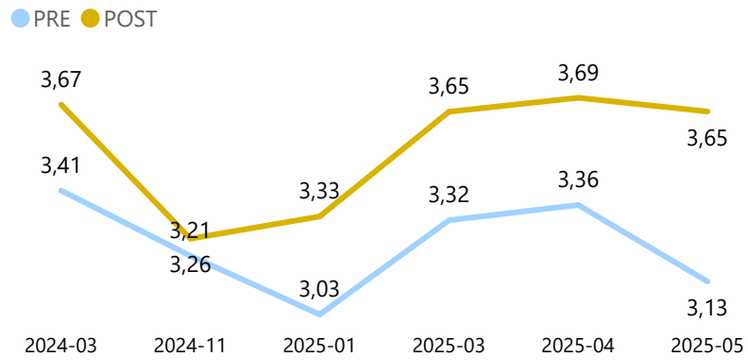


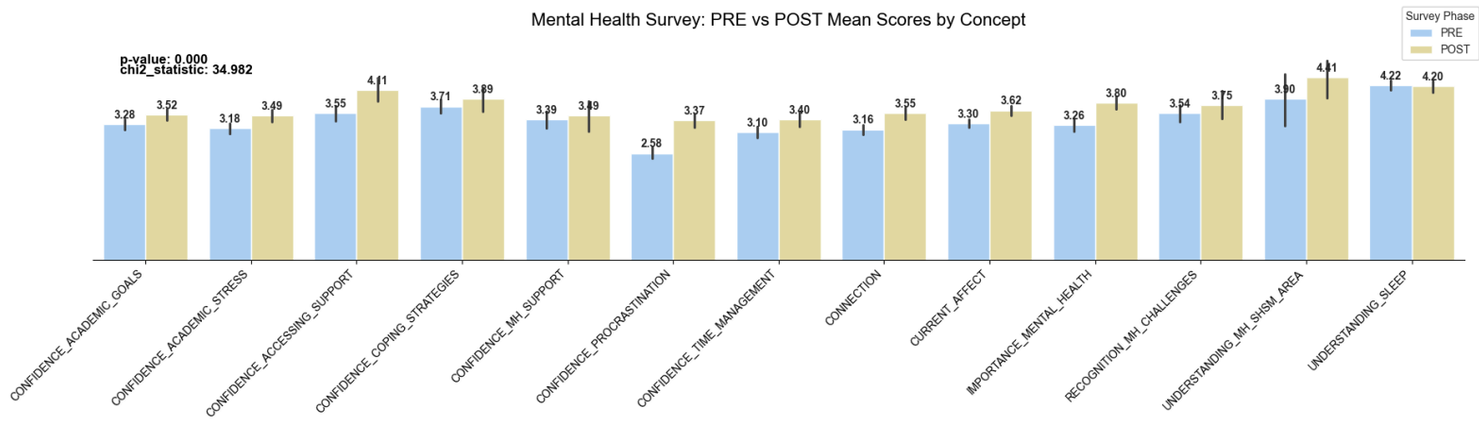
Year: All
Month: All
Hour: All
Phase: All
Concept: All
Survey: All
School: All
Questions: All

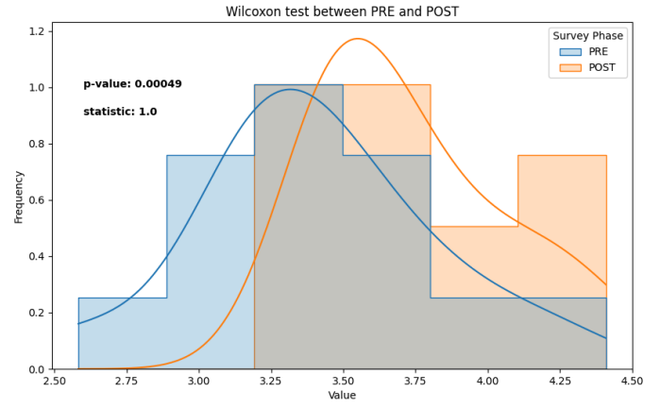
Frequency of Schools by Phase



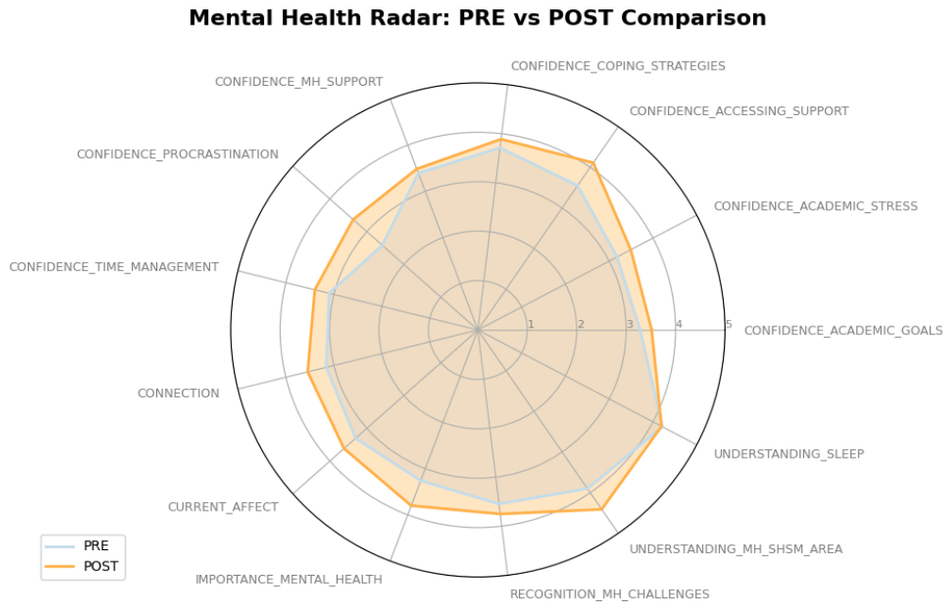
Survey Phase Trends

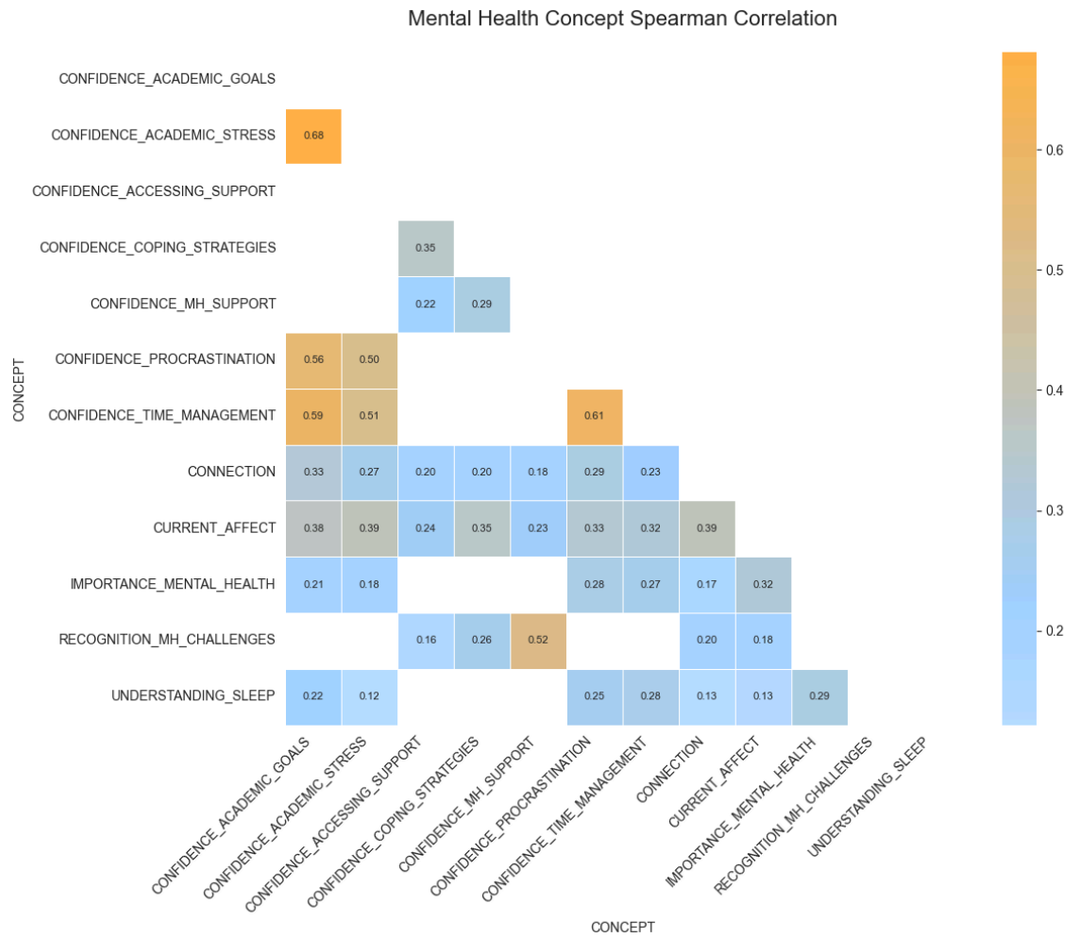




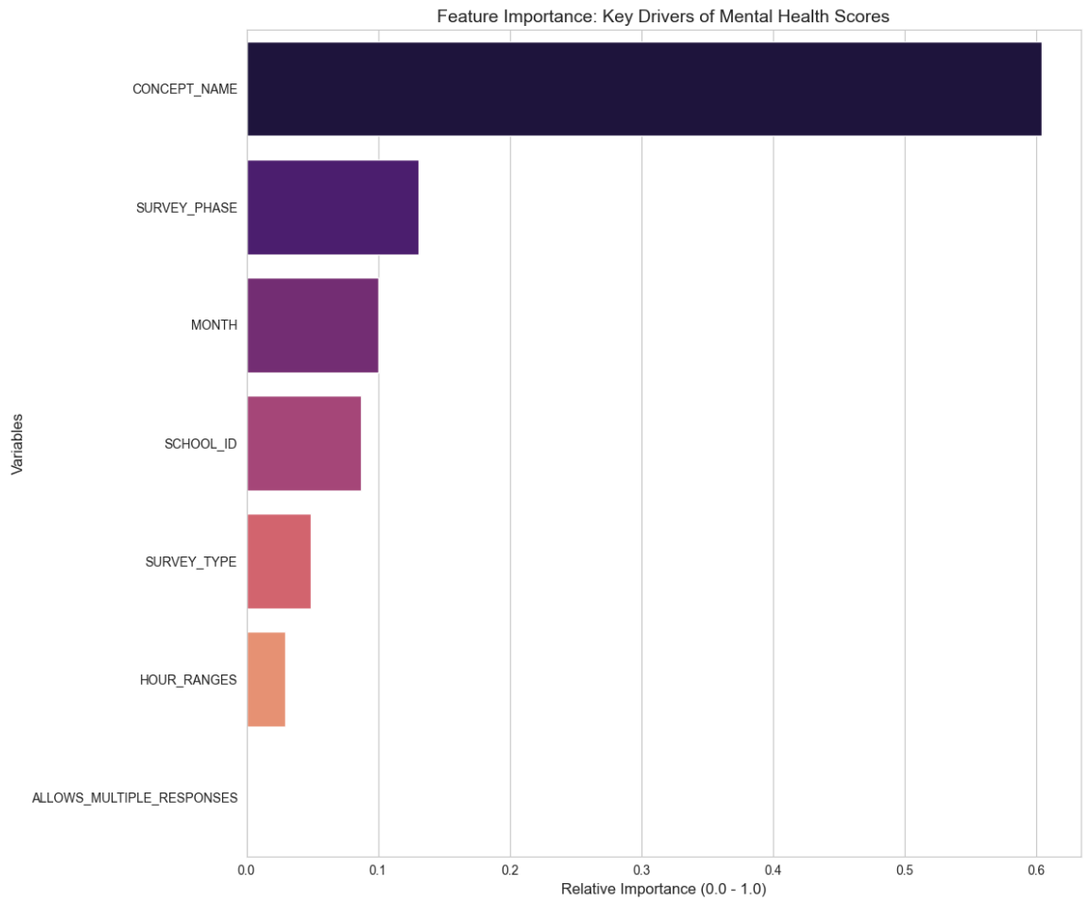


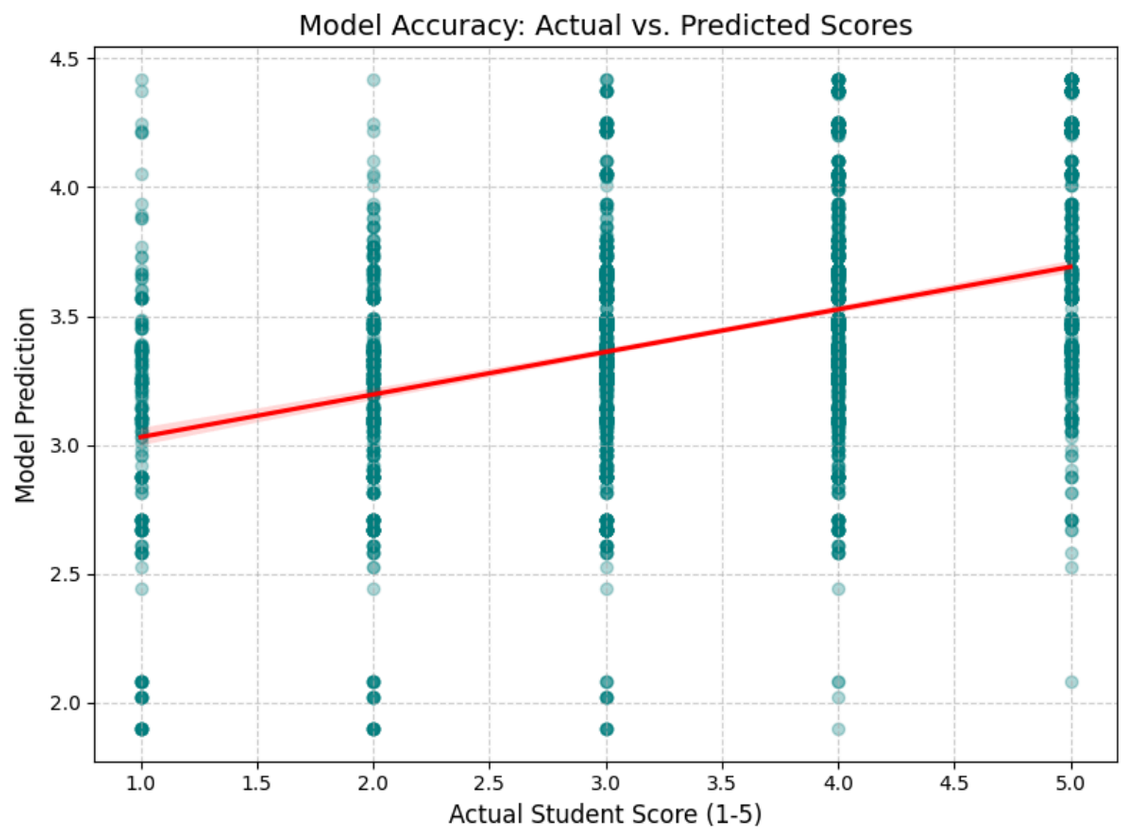
Year	Month	Hour	Phase	Concept	Survey	School	Questions
All	All	All	All	All	All	All	All





Year	Month	Hour	Phase	Concept	Survey	School	Questions
All	All	All	All	All	All	All	All





1.- **Universal Treatment Efficacy**
The primary conclusion is the **consistent positive shift** in student well-being across all measured metrics. The **Wilcoxon Signed-Rank test** ($p<0.05$) and the **bi-histogram** distributions confirm that the growth from PRE to POST phases is not due to random chance. This "Dominance Pattern" is visible across every School ID, proving that the intervention is robust and successfully overcomes local environmental variables.

2.- **Curriculum-Driven Growth**
The **Random Forest Feature Importance** and **Chi-Square Tests** identify the **Concept** (the specific skill taught) as the leading driver of student outcomes. With the intervention phase ranking as the second most influential variable, the data proves that the curriculum—particularly modules like **Connection** and **Procrastination** is the direct cause of the recorded improvements, rather than external seasonal factors.

3.- **The "Resilience Ecosystem"**
The **Spearman Correlation** analysis ($r=0.68$) reveals that student skills are deeply interconnected. Strong links between **Academic Goals and Stress Regulation** suggest a "compounding effect": when the program improves a student's goal-setting abilities, it simultaneously lowers their reported stress. This justifies a holistic approach to mental health rather than treating symptoms in isolation.

4.- **Scalability and Predictability**
With a **Mean Absolute Error (MAE) of 0.78**, the predictive model demonstrates high stability for social science data. The model's ability to account for the "Maturity Curve" where the gap between PRE and POST widening over time shows that the program is not just a temporary fix but a sustainable framework for building long-term student resilience across the district.

Response Key	Year - Month	Hours Ranges	Concept	School	Scale Type	Scale Levels	Measurement Levels	Allows Multiple Responses
R00001	2025-05	(08 - 12]	CURRENT_AFFECT	05	Likert Scale Numeric	{1: 'Terrible- worst day ever!', 5: 'Amazing- best day ever!'}	Ordinal	False
R00002	2025-05	(08 - 12]	CONNECTION	05	Likert Scale Numeric	{1: 'Not connected at all', 5: 'Extremely connected'}	Ordinal	False
R00003	2025-05	(08 - 12]	CONFIDENCE_TIME_MANAGEMENT	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Very confident'}	Ordinal	False
R00004	2025-05	(08 - 12]	CONFIDENCE_ACADEMIC_STRESS	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00005	2025-05	(08 - 12]	IMPORTANCE_MENTAL_HEALTH	05	Likert Scale Numeric		Ordinal	False
R00006	2025-05	(08 - 12]	CONFIDENCE_ACADEMIC_GOALS	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00007	2025-05	(08 - 12]	CONFIDENCE_PROCRASTINATION	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00008	2025-05	(08 - 12]	UNDERSTANDING_SLEEP	05	Likert Scale Text		Ordinal	False
R00009	2025-05	(08 - 12]	CURRENT_AFFECT	05	Likert Scale Numeric	{1: 'Terrible- worst day ever!', 5: 'Amazing- best day ever!'}	Ordinal	False
R00010	2025-05	(08 - 12]	CONNECTION	05	Likert Scale Numeric	{1: 'Not connected at all', 5: 'Extremely connected'}	Ordinal	False
R00011	2025-05	(08 - 12]	CONFIDENCE_TIME_MANAGEMENT	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Very confident'}	Ordinal	False
R00012	2025-05	(08 - 12]	CONFIDENCE_ACADEMIC_STRESS	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00013	2025-05	(08 - 12]	IMPORTANCE_MENTAL_HEALTH	05	Likert Scale Numeric		Ordinal	False
R00014	2025-05	(08 - 12]	CONFIDENCE_ACADEMIC_GOALS	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00015	2025-05	(08 - 12]	CONFIDENCE_PROCRASTINATION	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
R00016	2025-05	(08 - 12]	UNDERSTANDING_SLEEP	05	Likert Scale Text		Ordinal	False
R00017	2025-05	(08 - 12]	CURRENT_AFFECT	05	Likert Scale Numeric	{1: 'Terrible- worst day ever!', 5: 'Amazing- best day ever!'}	Ordinal	False
R00018	2025-05	(08 - 12]	CONNECTION	05	Likert Scale Numeric	{1: 'Not connected at all', 5: 'Extremely connected'}	Ordinal	False
R00019	2025-05	(08 - 12]	CONFIDENCE_TIME_MANAGEMENT	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Very confident'}	Ordinal	False
R00020	2025-05	(08 - 12]	CONFIDENCE_ACADEMIC_STRESS	05	Likert Scale Numeric	{1: 'Not confident', 5: 'Extremely confident'}	Ordinal	False
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